# On Axiomatic Approaches to Intertwining Operator Algebras

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2. Intertwining Operator Algebras
   - Definition of Intertwining Operator Algebras
   - Examples of Intertwining Operator Algebras
   - Duality for Intertwining Operator Algebras
3. The Jacobi identity for Intertwining Operator Algebras
   - Formulation of the Jacobi identity
   - Duality properties and the Jacobi identity
4. Genus-zero Moore-Seiberg equations
5. $S_3$-symmetry of the Jacobi identity
6. References

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- The direct sum of all inequivalent irreducible modules for a suitable vertex operator algebra, equipped with intertwining operators, has a natural algebraic structure called intertwining operator algebra, which is a natural generalization of the definition of vertex operator algebras.
- In the special case that the fusion rules are structure constants of group algebras of abelian groups, a notion of abelian intertwining operator algebra was introduced by Dong and Lepowsky, and examples were constructed.

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- Huang gave several general definitions of intertwining operator algebras, which are direct sums of modules (not necessarily irreducible) for vertex operator algebras equipped with intertwining operators among these modules satisfying some basic properties for intertwining operators.
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- We want to study the properties of general intertwining operator algebras, in analogy with the main properties of vertex operator algebras.
- We constructed fusing and braiding isomorphisms for intertwining operator algebras in the general case and proved that they satisfy the genus-zero Moore-Seiberg equations.
- We studied the duality properties of intertwining operator algebras and proved various equivalence results between axioms and properties.
- We proved an $S_3$-symmetry of the Jacobi identity for intertwining operator algebras, using the skew-symmetry property and the genus-zero Moore-Seiberg equations.

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# Definition of Intertwining Operator Algebras

An *intertwining operator algebra of central charge* $c \in \mathbb{C}$ consists of the following data:

- A vector space
$$W = \coprod_{a \in \mathcal{A}} W^a$$
graded by a finite set $\mathcal{A}$ containing a special element $e$ (graded by *color*).
- A vertex operator algebra structure of central charge $c$ on $W^e$, and a $W^e$-module structure on $W^a$ for each $a \in \mathcal{A}$.
- A subspace $\mathcal{V}_{a_1 a_2}^{a_3}$ of the space of all intertwining operators of type $\binom{W^{a_3}}{W^{a_1} W^{a_2}}$ for each triple $a_1, a_2, a_3 \in \mathcal{A}$, with its dimension denoted by $\mathcal{N}_{a_1 a_2}^{a_3}$.

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# Definition of Intertwining Operator Algebras

These data satisfy the following axioms for any $a_1, a_2, a_3, a_4, a_5, a_6 \in \mathcal{A}$, $w_{(a_i)} \in W^{a_i}$, $i = 1, 2, 3$, and $w'_{(a_4)} \in (W^{a_4})'$:

- The $W^e$-module structure on $W^e$ is the adjoint module structure. For any $a \in \mathcal{A}$, the space $\mathcal{V}_{ea}^{a}$ is the one-dimensional vector space spanned by the vertex operator for the $W^e$-module $W^a$. For any $a_1, a_2 \in \mathcal{A}$ such that $a_1 \neq a_2$, $\mathcal{V}_{ea_1}^{a_2} = 0$.

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# Definition of Intertwining Operator Algebras

- *Weight condition*: For any $a \in \mathcal{A}$ and the corresponding module $W^a = \coprod_{n \in \mathbb{C}} W^a_{(n)}$ graded by the action of $L(0)$, there exists $h_a \in \mathbb{R}$ such that $W^a_{(n)} = 0$ for $n \notin h_a + \mathbb{Z}$.

# Definition of Intertwining Operator Algebras

- *Convergence properties*:
  For any $m \in \mathbb{Z}_+$, $a_i, b_j \in \mathcal{A}$, $w_{(a_i)} \in W^{a_i}$, $\mathcal{Y}_i \in \mathcal{V}_{a_i\ b_{i+1}}^{b_i}$, $i = 1, \dots, m$, $j = 1, \dots, m+1$, $w'_{(b_1)} \in (W^{b_1})'$ and $w_{(b_{m+1})} \in W^{b_{m+1}}$, the series

  $$\langle w'_{(b_1)}, \mathcal{Y}_1(w_{(a_1)}, x_1) \cdots \mathcal{Y}_m(w_{(a_m)}, x_m) w_{(b_{m+1})} \rangle_{W^{b_1}} \Big|_{\substack{x_i^n = e^{n \log z_i}, \\ i=1,\dots,m,\ n \in \mathbb{R}}}$$

  is absolutely convergent when $|z_1| > \cdots > |z_m| > 0$ and its sum can be analytically extended to a multivalued analytic function on the region given by $z_i \neq 0$, $i = 1, \dots, m$, $z_i \neq z_j$, $i \neq j$.

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$$\langle w'_{(a_4)}, \mathcal{Y}_2(\mathcal{Y}_1(w_{(a_1)}, x_0)w_{(a_2)}, x_2)w_{(a_3)}\rangle w^{a_4}\Big|_{\substack{x_0^n = e^{n\log(z_1-z_2)},\\ x_2^n = e^{n\log z_2},\ n\in\mathbb{R}}}$$

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- *Associativity*:
  For any $\mathcal{Y}_1 \in \mathcal{V}_{a_1 a_5}^{a_4}$ and $\mathcal{Y}_2 \in \mathcal{V}_{a_2 a_3}^{a_5}$, there exist $\mathcal{Y}_{3,i}^{a} \in \mathcal{V}_{a_1 a_2}^{a}$ and $\mathcal{Y}_{4,i}^{a} \in \mathcal{V}_{a a_3}^{a_4}$ for $i = 1, \dots, \mathcal{N}_{a_1 a_2}^{a} \mathcal{N}_{a a_3}^{a_4}$ and $a \in \mathcal{A}$, such that the (multivalued) analytic function

$$\langle w'_{(a_4)}, \mathcal{Y}_1(w_{(a_1)}, x_1)\mathcal{Y}_2(w_{(a_2)}, x_2)w_{(a_3)}\rangle W^{a_4}\big|_{x_1=z_1, x_2=z_2}$$

  defined in the region $|z_1| > |z_2| > 0$ and the (multivalued) analytic function

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# Definition of Intertwining Operator Algebras

- *Skew-symmetry*:
  The restriction of $\Omega_{-1}$ to $\mathcal{V}_{a_1 a_2}^{a_3}$ is an isomorphism from $\mathcal{V}_{a_1 a_2}^{a_3}$ to $\mathcal{V}_{a_2 a_1}^{a_3}$.

  (Note 1. $\Omega_{-1}$ is a linear map which maps intertwining operator of type $\binom{W^{a_3}}{W^{a_1} W^{a_2}}$ to intertwining operator of type $\binom{W^{a_3}}{W^{a_2} W^{a_1}}$ for any $a_1, a_2, a_3 \in \mathcal{A}$. Actually, for any $r \in \mathbb{Z}$, we define

  $$\Omega_r(\mathcal{Y}) : W^{a_2} \otimes W^{a_1} \to W^{a_3}\{x\}$$

  by

  $$\Omega_r(\mathcal{Y})(w_{(a_2)}, x) w_{(a_1)} = e^{xL(-1)} \mathcal{Y}(w_{(a_1)}, e^{(2r+1)\pi i} x) w_{(a_2)}$$

  for any $a_1, a_2, a_3 \in \mathcal{A}$, where $w_{(a_1)} \in W^{a_1}$, $w_{(a_2)} \in W^{a_2}$ and $\mathcal{Y}$ is an intertwining operator of type $\binom{W^{a_3}}{W^{a_1} W^{a_2}}$.)
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  C. Dong and J. Lepowsky, Generalized Vertex Algebras and Relative Vertex Operators, *Progress in Math.*, Vol. 112. Birkhauser, Boston, 1993. )

- The direct sum of all inequivalent irreducible modules for the vertex operator algebra associated to an affine Lie algebra at positive level forms an intertwining operator algebra.

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# Duality for Intertwining Operator Algebras

Second Associativity:
Let $(W, \mathcal{A}, \{\mathcal{V}_{a_1 a_2}^{a_3}\}, \mathbf{1}, \omega)$ be an intertwining operator algebra.
Then we have the following *second associativity*:
For any $a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}_{a_1 a_2}^{a_5}$ and $\mathcal{Y}_2 \in \mathcal{V}_{a_5 a_3}^{a_4}$, there exist $\mathcal{Y}_{3,j}^{a} \in \mathcal{V}_{a_1 a}^{a_4}$ and $\mathcal{Y}_{4,j}^{a} \in \mathcal{V}_{a_2 a_3}^{a}$ for $j = 1, \ldots, \mathcal{N}_{a_1 a}^{a_4} \mathcal{N}_{a_2 a_3}^{a}$ and $a \in \mathcal{A}$, such that for $w_{(a_i)} \in W^{a_i}$, $i = 1, 2, 3$, and $w'_{(a_4)} \in (W^{a_4})'$, the (multivalued) analytic function

$$\langle w'_{(a_4)}, \mathcal{Y}_2(\mathcal{Y}_1(w_{(a_1)}, x_0) w_{(a_2)}, x_2) w_{(a_3)} \rangle W^{a_4} |_{x_0 = z_1 - z_2, x_2 = z_2}$$

defined in the region $|z_2| > |z_1 - z_2| > 0$

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defined in the region $|z_2| > |z_1 - z_2| > 0$

# Duality for Intertwining Operator Algebras

Second Associativity:
and the (multivalued) analytic function

$$\sum_{a\in\mathcal{A}}\sum_{j=1}^{\mathcal{N}_{a_1 a}^{a_4}\mathcal{N}_{a_2 a_3}^{a}}\langle w'_{(a_4)}, \mathcal{Y}_{3,j}^{a}(w_{(a_1)}, x_1)\mathcal{Y}_{4,j}^{a}(w_{(a_2)}, x_2)w_{(a_3)}\rangle W^{a_4}\Big|_{x_1=z_1, x_2=z_2}$$

defined in the region $|z_1| > |z_2| > 0$ are equal on the intersection $|z_1| > |z_2| > |z_1 - z_2| > 0$.

# Duality for Intertwining Operator Algebras

Commutativity:
Let $(W, \mathcal{A}, \{\mathcal{V}_{a_1 a_2}^{a_3}\}, \mathbf{1}, \omega)$ be an intertwining operator algebra.
Then we have the following *commutativity*:
For any $a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}_{a_1 a_5}^{a_4}$ and $\mathcal{Y}_2 \in \mathcal{V}_{a_2 a_3}^{a_5}$, there exist $\mathcal{Y}_{5,j}^{a} \in \mathcal{V}_{a_2 a}^{a_4}$ and $\mathcal{Y}_{6,j}^{a} \in \mathcal{V}_{a_1 a_3}^{a}$ for $j = 1, \ldots, \mathcal{N}_{a_2 a}^{a_4} \mathcal{N}_{a_1 a_3}^{a}$ and $a \in \mathcal{A}$, such that for $w_{(a_i)} \in W^{a_i}$, $i = 1, 2, 3$, and $w'_{(a_4)} \in (W^{a_4})'$, the (multivalued) analytic function

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# Duality for Intertwining Operator Algebras

Commutativity:
and the (multivalued) analytic function

$$\sum_{a\in\mathcal{A}} \sum_{j=1}^{\mathcal{N}_{a_2 a}^{a_4}\mathcal{N}_{a_1 a_3}^{a}} \langle w'_{(a_4)}, \mathcal{Y}_{5,j}^{a}(w_{(a_2)}, x_2)\mathcal{Y}_{6,j}^{a}(w_{(a_1)}, x_1)w_{(a_3)}\rangle w^{a_4}\Big|_{x_1=z_1, x_2=z_2}$$

defined in the region $|z_2| > |z_1| > 0$ are analytic extensions of each other.

# Some consequences

**Theorem**

*In the presence of the axioms for intertwining operator algebras except for associativity and skew-symmetry, the associativity property of intertwining operator algebras follows from commutativity and skew-symmetry.*

**Theorem**

*In the presence of the axioms for intertwining operator algebras except for skew-symmetry, we assume that commutativity holds, and that the restriction of $\Omega$ to $\mathcal{V}_{ea}^{b}$ is an isomorphism from $\mathcal{V}_{ea}^{b}$ to $\mathcal{V}_{ae}^{b}$ for any $a, b \in \mathcal{A}$. Then the restriction of $\Omega$ to $\mathcal{V}_{a_1 a_2}^{a_3}$ is an isomorphism from $\mathcal{V}_{a_1 a_2}^{a_3}$ to $\mathcal{V}_{a_2 a_1}^{a_3}$ for any $a_1, a_2, a_3 \in \mathcal{A}$.*

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# On Axiomatic Approaches to Intertwining Operator Algebras

1. Backgrounds
2. Intertwining Operator Algebras
   - Definition of Intertwining Operator Algebras
   - Examples of Intertwining Operator Algebras
   - Duality for Intertwining Operator Algebras
3. The Jacobi identity for Intertwining Operator Algebras
   - Formulation of the Jacobi identity
   - Duality properties and the Jacobi identity
4. Genus-zero Moore-Seiberg equations
5. $S_3$-symmetry of the Jacobi identity
6. References

# Multiplication of intertwining operators

In correspondence to the multiplication of intertwining operators, we define a map

$$\begin{aligned}\mathbf{P}: \coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5} &\to (\mathrm{Hom}(W\otimes W\otimes W, W))\{x_1,x_2\}\\ \mathcal{Z} &\mapsto \mathbf{P}(\mathcal{Z})\end{aligned}$$

by linearity and by

$$\begin{aligned}&(\mathbf{P}(\mathcal{Y}_1\otimes\mathcal{Y}_2))(w_{(a_6)}, w_{(a_7)}, w_{(a_8)}; x_1, x_2)\\ &\quad= \begin{cases}\mathcal{Y}_1(w_{(a_6)}, x_1)\mathcal{Y}_2(w_{(a_7)}, x_2)w_{(a_8)}, & a_6=a_1, a_7=a_2, a_8=a_3,\\ 0, & \text{otherwise}\end{cases}\end{aligned}$$

for $a_1,\dots,a_8\in\mathcal{A}$, $\mathcal{Y}_1\in\mathcal{V}_{a_1a_5}^{a_4}$, $\mathcal{Y}_2\in\mathcal{V}_{a_2a_3}^{a_5}$, and $w_{(a_6)}\in W^{a_6}$, $w_{(a_7)}\in W^{a_7}$, $w_{(a_8)}\in W^{a_8}$,

# Multiplication of intertwining operators

from which we get an isomorphism

$$\tilde{\mathbf{P}} : \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}}{\operatorname{Ker} \mathbf{P}} \longrightarrow \mathbf{P}\left(\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}\right)$$

# Multiplication of intertwining operators

which makes the following diagram commute:

$$\begin{tikzcd}
\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5} \arrow[r, "\mathbf{P}"] \arrow[d, "\pi_P"'] & \mathbf{P}\left(\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5}\right), \\
\dfrac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5}}{\operatorname{Ker}\mathbf{P}} \arrow[ur, "\tilde{\mathbf{P}}"] &
\end{tikzcd}$$

where $\pi_P$ is the corresponding canonical projective map.

# Iterates of intertwining operators

In correspondence to the iterates of intertwining operators, we define a map

$$\begin{aligned} \mathbf{I} : \coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4} &\to (\mathrm{Hom}(W\otimes W\otimes W, W))\{x_0,x_2\} \\ \mathcal{Z} &\mapsto \mathbf{I}(\mathcal{Z}) \end{aligned}$$

by linearity and by

$$\begin{aligned} &(\mathbf{I}(\mathcal{Y}_1\otimes\mathcal{Y}_2))(w_{(a_6)}, w_{(a_7)}, w_{(a_8)}; x_0, x_2) \\ &\quad = \begin{cases} \mathcal{Y}_2(\mathcal{Y}_1(w_{(a_6)}, x_0)w_{(a_7)}, x_2)w_{(a_8)}, & a_6 = a_1, a_7 = a_2, a_8 = a_3, \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

for $a_1,\dots,a_8\in\mathcal{A}$, $\mathcal{Y}_1\in\mathcal{V}_{a_1a_2}^{a_5}$, $\mathcal{Y}_2\in\mathcal{V}_{a_5a_3}^{a_4}$, and $w_{(a_6)}\in W^{a_6}$, $w_{(a_7)}\in W^{a_7}$, $w_{(a_8)}\in W^{a_8}$,

# Iterates of intertwining operators

from which we get an isomorphism

$$\tilde{\mathbf{I}}: \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\operatorname{Ker}\mathbf{I}} \longrightarrow \mathbf{I}\left(\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}\right),$$

# Iterates of intertwining operators

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$$\begin{tikzcd}
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\dfrac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\operatorname{Ker}\mathcal{I}} \arrow[ur, "\tilde{\mathcal{I}}"] &
\end{tikzcd}$$

where $\pi_I$ is the corresponding canonical projective map.

# Fusing isomorphism

From the associativity of intertwining operator algebras, we obtain a *fusing isomorphism*:

$$\mathcal{F}: \quad \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5}}{\mathrm{Ker}\ \mathbf{P}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\mathrm{Ker}\ \mathbf{I}}$$

determined by linearity and by

$$\mathcal{F}(\mathcal{Y}_1\otimes\mathcal{Y}_2+\mathrm{Ker}\ \mathbf{P}) = \sum_{a\in\mathcal{A}}\sum_{i=1}^{\mathcal{N}_{a_1a_2}^{a}\mathcal{N}_{aa_3}^{a_4}} \mathcal{Y}_{3,i}^{a}\otimes\mathcal{Y}_{4,i}^{a}+\mathrm{Ker}\ \mathbf{I}$$

for $a_1,\cdots,a_5\in\mathcal{A}$, $\mathcal{Y}_1\in\mathcal{V}_{a_1a_5}^{a_4}$ and $\mathcal{Y}_2\in\mathcal{V}_{a_2a_3}^{a_5}$,

# Fusing isomorphism

where $\{\mathcal{Y}_{3,i}^{a} \in \mathcal{V}_{a_1 a_2}^{a}, \mathcal{Y}_{4,i}^{a} \in \mathcal{V}_{a a_3}^{a_4} \mid i = 1, \cdots, \mathcal{N}_{a_1 a_2}^{a} \mathcal{N}_{a a_3}^{a_4}, a \in \mathcal{A}\}$ is a set of intertwining operators satisfying that for any $w_1, w_2, w_3 \in W$ and $w' \in W'$,

$$\sum_{a \in \mathcal{A}} \sum_{i=1}^{\mathcal{N}_{a_1 a_2}^{a} \mathcal{N}_{a a_3}^{a_4}} \langle w', (\mathbf{I}(\mathcal{Y}_{3,i}^{a} \otimes \mathcal{Y}_{4,i}^{a}))(w_1, w_2, w_3; x_0, x_2) \rangle_W \Big|_{\substack{x_0^n = e^{n \log(z_1 - z_2)} \\ x_2^n = e^{n \log z_2}}}$$

is equal to

$$\langle w', (\mathbf{P}(\mathcal{Y}_1 \otimes \mathcal{Y}_2))(w_1, w_2, w_3; x_1, x_2) \rangle_W |_{x_1^n = e^{n \log z_1}, x_2^n = e^{n \log z_2}}$$

in the region

$$\{(z_1, z_2) \in \mathbb{C}^2 \mid \mathrm{Re} z_1 > \mathrm{Re} z_2 > \mathrm{Re}(z_1 - z_2) > 0, \\ \mathrm{Im} z_1 > \mathrm{Im} z_2 > \mathrm{Im}(z_1 - z_2) > 0\}.$$

# Skew-symmetry isomorphisms

From $\Omega$ and its inverse, we obtain several isomorphisms:

$\tilde{\Omega}^{(1)}, (\widetilde{\Omega^{-1}})^{(1)}$ :

$$\frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\mathrm{Ker}\ \mathbf{I}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_2a_1}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\mathrm{Ker}\ \mathbf{I}}$$

$\tilde{\Omega}^{(2)}, (\widetilde{\Omega^{-1}})^{(2)}$ :

$$\frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5}}{\mathrm{Ker}\ \mathbf{P}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_2a_3}^{a_5}\otimes\mathcal{V}_{a_5a_1}^{a_4}}{\mathrm{Ker}\ \mathbf{I}}$$

# Skew-symmetry isomorphisms

$\tilde{\Omega}^{(3)}, (\widetilde{\Omega^{-1}})^{(3)}$ :

$$\frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5}\otimes\mathcal{V}_{a_5a_3}^{a_4}}{\mathrm{Ker}\ \mathbf{I}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_3a_5}^{a_4}\otimes\mathcal{V}_{a_1a_2}^{a_5}}{\mathrm{Ker}\ \mathbf{P}}$$

$\tilde{\Omega}^{(4)}, (\widetilde{\Omega^{-1}})^{(4)}$ :

$$\frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_2a_3}^{a_5}}{\mathrm{Ker}\ \mathbf{P}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4}\otimes\mathcal{V}_{a_3a_2}^{a_5}}{\mathrm{Ker}\ \mathbf{P}}$$

# Skew-symmetry isomorphisms

These skew-symmetry isomorphisms have relations:

$$(\tilde{\Omega}^{(2)})^{-1} = (\widetilde{\Omega^{-1}})^{(3)}, \quad ((\widetilde{\Omega^{-1}})^{(2)})^{-1} = \tilde{\Omega}^{(3)},$$

$$(\tilde{\Omega}^{(1)})^{-1} = (\widetilde{\Omega^{-1}})^{(1)}, \quad (\tilde{\Omega}^{(4)})^{-1} = (\widetilde{\Omega^{-1}})^{(4)}.$$

# Braiding isomorphism

Using the fusing isomorphism and the isomorphism $\tilde{\Omega}^{(1)}$, we deduce a *braiding isomorphism*

$$\mathcal{B} = \mathcal{F}^{-1} \circ \tilde{\Omega}^{(1)} \circ \mathcal{F} : \frac{\coprod_{a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}}{\operatorname{Ker} \mathbf{P}} \longrightarrow \frac{\coprod_{a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}} \mathcal{V}_{a_2 a_5}^{a_4} \otimes \mathcal{V}_{a_1 a_3}^{a_5}}{\operatorname{Ker} \mathbf{P}}.$$

# Vector Space of Special Multivalued Analytic Functions

Consider the following simply connected regions in $\mathbb{C}^2$:

$R_1$: obtained by cutting the region $|z_1| > |z_2| > 0$ along the intersection of this region with $\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,

$R_2$: obtained by cutting the region $|z_2| > |z_1| > 0$ along the intersection of this region with $\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,

$R_3$: obtained by cutting the region $|z_2| > |z_1 - z_2| > 0$ along the intersection of this region with $\{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 - z_2 \in [0, +\infty)\}$.

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$R_2$: obtained by cutting the region $|z_2| > |z_1| > 0$ along the intersection of this region with
$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,
$R_3$: obtained by cutting the region $|z_2| > |z_1 - z_2| > 0$ along the intersection of this region with
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$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,
$R_3$: obtained by cutting the region $|z_2| > |z_1 - z_2| > 0$ along the intersection of this region with
$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 - z_2 \in [0, +\infty)\}$.

# Vector Space of Special Multivalued Analytic Functions

Consider the following simply connected regions in $\mathbb{C}^2$:
$R_1$: obtained by cutting the region $|z_1| > |z_2| > 0$ along the intersection of this region with
$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,
$R_2$: obtained by cutting the region $|z_2| > |z_1| > 0$ along the intersection of this region with
$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\}$,
$R_3$: obtained by cutting the region $|z_2| > |z_1 - z_2| > 0$ along the intersection of this region with
$\{(z_1, z_2) \in \mathbb{C}^2 \mid z_2 \in [0, +\infty)\} \cup \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1 - z_2 \in [0, +\infty)\}$.

# Vector Space of Special Multivalued Analytic Functions

For $a_1, a_2, a_3, a_4 \in \mathcal{A}$, let $\mathbb{G}^{a_1,a_2,a_3,a_4}$ be the set of multivalued analytic functions on

$$M^2 = \{(z_1, z_2) \in \mathbb{C}^2 \mid z_1, z_2 \neq 0, z_1 \neq z_2\}$$

with a choice of a single-valued branch in the region $R_1$ satisfying the following property: In the regions $|z_1| > |z_2| > 0$, $|z_2| > |z_1| > 0$ and $|z_2| > |z_1 - z_2| > 0$, any branch of $f(z_1, z_2) \in \mathbb{G}^{a_1,a_2,a_3,a_4}$ can be expanded as

# Vector Space of Special Multivalued Analytic Functions

$$\sum_{a\in\mathcal{A}} z_1^{h_{a_4}-h_{a_1}-h_a} z_2^{h_a-h_{a_2}-h_{a_3}} F_a(z_1, z_2),$$
$$\sum_{a\in\mathcal{A}} z_2^{h_{a_4}-h_{a_2}-h_a} z_1^{h_a-h_{a_1}-h_{a_3}} G_a(z_1, z_2)$$

and

$$\sum_{a\in\mathcal{A}} z_2^{h_{a_4}-h_a-h_{a_3}} (z_1 - z_2)^{h_a-h_{a_1}-h_{a_2}} H_a(z_1, z_2),$$

respectively,

# Vector Space of Special Multivalued Analytic Functions

where for $a \in \mathcal{A}$,

$$F_a(z_1, z_2) \in \mathbb{C}[[z_2/z_1]][z_1, z_1^{-1}, z_2, z_2^{-1}],$$

$$G_a(z_1, z_2) \in \mathbb{C}[[z_1/z_2]][z_1, z_1^{-1}, z_2, z_2^{-1}]$$

and

$$H_a(z_1, z_2) \in \mathbb{C}[[(z_1 - z_2)/z_2]][z_2, z_2^{-1}, z_1 - z_2, (z_1 - z_2)^{-1}].$$

# Vector Space of Special Multivalued Analytic Functions

We call the chosen single-valued branch on $R_1$ of an element of $\mathbb{G}^{a_1,a_2,a_3,a_4}$ the *preferred branch on* $R_1$.

For any function in $\mathbb{G}^{a_1,a_2,a_3,a_4}$, we can obtain single-valued branches on $R_2$ and $R_3$ with a particular fixed extension path from its preferred branch on $R_1$. We call these branches preferred branches on $R_2$ and $R_3$ respectively.

# Vector Space of Special Multivalued Analytic Functions

We call the chosen single-valued branch on $R_1$ of an element of $\mathbb{G}^{a_1,a_2,a_3,a_4}$ the *preferred branch on* $R_1$.
For any function in $\mathbb{G}^{a_1,a_2,a_3,a_4}$, we can obtain single-valued branches on $R_2$ and $R_3$ with a particular fixed extension path from its preferred branch on $R_1$. We call these branches preferred branches on $R_2$ and $R_3$ respectively.

# Vector Space of Special Multivalued Analytic Functions

Given an element of $\mathbb{G}^{a_1,a_2,a_3,a_4}$, the preferred branches of this function on $R_1$, $R_2$ and $R_3$ give formal series in

$$\coprod_{a\in\mathcal{A}} x_1^{h_{a_4}-h_{a_1}-h_a} x_2^{h_a-h_{a_2}-h_{a_3}} \mathbb{C}[[x_2/x_1]][x_1, x_1^{-1}, x_2, x_2^{-1}],$$

$$\coprod_{a\in\mathcal{A}} x_2^{h_{a_4}-h_{a_2}-h_a} x_1^{h_a-h_{a_1}-h_{a_3}} \mathbb{C}[[x_1/x_2]][x_1, x_1^{-1}, x_2, x_2^{-1}]$$

and

$$\coprod_{a\in\mathcal{A}} x_2^{h_{a_4}-h_a-h_{a_3}} x_0^{h_a-h_{a_1}-h_{a_2}} \mathbb{C}[[x_0/x_2]][x_0, x_0^{-1}, x_2, x_2^{-1}],$$

respectively.

# Vector Space of Special Multivalued Analytic Functions

Thus we have linear maps

$$\begin{aligned}
\iota_{12}: \quad & \mathbb{G}^{a_1,a_2,a_3,a_4} \\
& \to \coprod_{a\in\mathcal{A}} x_1^{h_{a_4}-h_{a_1}-h_a} x_2^{h_a-h_{a_2}-h_{a_3}} \mathbb{C}[[x_2/x_1]][x_1, x_1^{-1}, x_2, x_2^{-1}] \\
\iota_{21}: \quad & \mathbb{G}^{a_1,a_2,a_3,a_4} \\
& \to \coprod_{a\in\mathcal{A}} x_2^{h_{a_4}-h_{a_2}-h_a} x_1^{h_a-h_{a_1}-h_{a_3}} \mathbb{C}[[x_1/x_2]][x_1, x_1^{-1}, x_2, x_2^{-1}] \\
\iota_{20}: \quad & \mathbb{G}^{a_1,a_2,a_3,a_4} \\
& \to \coprod_{a\in\mathcal{A}} x_2^{h_{a_4}-h_a-h_{a_3}} x_0^{h_a-h_{a_1}-h_{a_2}} \mathbb{C}[[x_0/x_2]][x_0, x_0^{-1}, x_2, x_2^{-1}]
\end{aligned}$$

Since analytic extensions are unique, these maps are injective.

# Vector Space of Special Multivalued Analytic Functions

For any $a_1, a_2, a_3, a_4 \in \mathcal{A}$, $\mathbb{G}^{a_1,a_2,a_3,a_4}$ is a module over the ring

$$\mathbb{C}[x_1, x_1^{-1}, x_2, x_2^{-1}, (x_1 - x_2)^{-1}].$$

We have the following lemma proved in [H]:

**Lemma**

*For any $a_1, a_2, a_3, a_4 \in \mathcal{A}$, the module $\mathbb{G}^{a_1,a_2,a_3,a_4}$ is free.*

We fix a basis $\{e_\alpha^{a_1,a_2,a_3,a_4}\}_{\alpha\in\mathbb{A}(a_1,a_2,a_3,a_4)}$ of the free module $\mathbb{G}^{a_1,a_2,a_3,a_4}$ for any $a_1, a_2, a_3, a_4 \in \mathcal{A}$.

# Jacobi identity

For any $a_1, a_2, a_3, a_4 \in \mathcal{A}$, there exist linear maps

$$\begin{aligned} f_\alpha^{a_1,a_2,a_3,a_4} : W^{a_1} \otimes W^{a_2} \otimes W^{a_3} \otimes \pi_P\Big(\coprod_{a_5 \in \mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5}\Big) \\ \to W^{a_4}[[x_2/x_1]][x_1, x_1^{-1}, x_2, x_2^{-1}] \\ w_{(a_1)} \otimes w_{(a_2)} \otimes w_{(a_3)} \otimes [\mathcal{Z}]_P \\ \mapsto f_\alpha^{a_1,a_2,a_3,a_4}\big(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_P; x_1, x_2\big), \end{aligned}$$

# Jacobi identity

$$g_\alpha^{a_1,a_2,a_3,a_4} : W^{a_1} \otimes W^{a_2} \otimes W^{a_3} \otimes \pi_P\Big(\coprod_{a_5 \in \mathcal{A}} \mathcal{V}_{a_2 a_5}^{a_4} \otimes \mathcal{V}_{a_1 a_3}^{a_5}\Big)$$

$$\to W^{a_4}[[x_1/x_2]][x_1, x_1^{-1}, x_2, x_2^{-1}]$$

$$w_{(a_1)} \otimes w_{(a_2)} \otimes w_{(a_3)} \otimes [\mathcal{Z}]_P$$

$$\mapsto g_\alpha^{a_1,a_2,a_3,a_4}\big(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_P; x_1, x_2\big)$$

# Jacobi identity

and

$$\begin{aligned} h_\alpha^{a_1,a_2,a_3,a_4} : W^{a_1} \otimes W^{a_2} \otimes W^{a_3} \otimes \pi_I\Big(\coprod_{a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5} \otimes \mathcal{V}_{a_5a_3}^{a_4}\Big) \\ \to W^{a_4}[[x_0/x_2]][x_0, x_0^{-1}, x_2, x_2^{-1}] \\ w_{(a_1)} \otimes w_{(a_2)} \otimes w_{(a_3)} \otimes [\mathcal{Z}]_I \\ \mapsto h_\alpha^{a_1,a_2,a_3,a_4}\big(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_I; x_0, x_2\big) \end{aligned}$$

for $\alpha \in \mathbb{A}(a_1, a_2, a_3, a_4)$,

# Jacobi identity

such that for any $w_{(a_1)} \in W^{a_1}$, $w_{(a_2)} \in W^{a_2}$, $w_{(a_3)} \in W^{a_3}$, and any

$$\mathcal{Z} \in \coprod_{a_5 \in \mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5} \subset \coprod_{a_1, a_2, a_3, a_4, a_5 \in \mathcal{A}} \mathcal{V}_{a_1 a_5}^{a_4} \otimes \mathcal{V}_{a_2 a_3}^{a_5},$$

only finitely many of

$$f_\alpha^{a_1, a_2, a_3, a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_P; x_1, x_2),$$

$$g_\alpha^{a_1, a_2, a_3, a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{B}([\mathcal{Z}]_P); x_1, x_2),$$

and

$$h_\alpha^{a_1, a_2, a_3, a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{F}([\mathcal{Z}]_P); x_0, x_2),$$

$\alpha \in \mathbb{A}(a_1, a_2, a_3, a_4)$, are nonzero,

# Jacobi identity

$$(\tilde{\mathbf{P}}([\mathcal{Z}]_P))(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}; x_1, x_2)$$
$$= \sum_{\alpha\in\mathbb{A}(a_1,a_2,a_3,a_4)} f_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_P; x_1, x_2)$$
$$\iota_{12}\left(e_\alpha^{a_1,a_2,a_3,a_4}\right),$$

$$(\tilde{\mathbf{P}}(\mathcal{B}([\mathcal{Z}]_P)))(w_{(a_2)}, w_{(a_1)}, w_{(a_3)}; x_2, x_1)$$
$$= \sum_{\alpha\in\mathbb{A}(a_1,a_2,a_3,a_4)} g_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{B}([\mathcal{Z}]_P); x_1, x_2)$$
$$\iota_{21}\left(e_\alpha^{a_1,a_2,a_3,a_4}\right),$$

# Jacobi identity

$$(\tilde{\mathbf{i}}(\mathcal{F}([\mathcal{Z}]_P)))(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}; x_0, x_2)$$
$$= \sum_{\alpha\in\mathbb{A}(a_1,a_2,a_3,a_4)} h_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{F}([\mathcal{Z}]_P); x_0, x_2)$$
$$\iota_{20}(e_\alpha^{a_1,a_2,a_3,a_4}),$$

and the following *Jacobi identity* holds:

$$x_0^{-1}\delta\left(\frac{x_1-x_2}{x_0}\right) f_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, [\mathcal{Z}]_P; x_1, x_2)$$
$$-x_0^{-1}\delta\left(\frac{x_2-x_1}{-x_0}\right) g_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{B}([\mathcal{Z}]_P); x_1, x_2)$$
$$= x_2^{-1}\delta\left(\frac{x_1-x_0}{x_2}\right) h_\alpha^{a_1,a_2,a_3,a_4}(w_{(a_1)}, w_{(a_2)}, w_{(a_3)}, \mathcal{F}([\mathcal{Z}]_P); x_0, x_2)$$

# Duality properties and the Jacobi identity

**Theorem**

*In the presence of the axioms for intertwining operator algebras except for the associativity and skew-symmetry properties, the commutativity and associativity properties follow from the Jacobi identity.*

# On Axiomatic Approaches to Intertwining Operator Algebras

1. Backgrounds
2. Intertwining Operator Algebras
   - Definition of Intertwining Operator Algebras
   - Examples of Intertwining Operator Algebras
   - Duality for Intertwining Operator Algebras
3. The Jacobi identity for Intertwining Operator Algebras
   - Formulation of the Jacobi identity
   - Duality properties and the Jacobi identity
4. Genus-zero Moore-Seiberg equations
5. $S_3$-symmetry of the Jacobi identity
6. References

# Genus-zero Moore-Seiberg equations

In correspondence to the multiplications, iterates and the mixtures of the two operations of three intertwining operators, we have five linear maps:
The first one is

$$\mathbf{PP}: \quad \coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_6}^{a_5}\otimes\mathcal{V}_{a_2a_7}^{a_6}\otimes\mathcal{V}_{a_3a_4}^{a_7} \longrightarrow (\mathrm{Hom}(W\otimes W\otimes W\otimes W, W))\{x_1,x_2,x_3\}$$

defined by linearity and by

$$(\mathbf{PP}(\mathcal{Y}_1 \otimes \mathcal{Y}_2 \otimes \mathcal{Y}_3))(w_{(a_{i_1})}, w_{(a_{i_2})}, w_{(a_{i_3})}, w_{(a_{i_4})}; x_1, x_2, x_3)$$
$$= \begin{cases} \mathcal{Y}_1(w_{(a_{i_1})}, x_1)\mathcal{Y}_2(w_{(a_{i_2})}, x_2)\mathcal{Y}_3(w_{(a_{i_3})}, x_3)w_{(a_{i_4})}, & i_1 = 1, i_2 = 2, \\ & i_3 = 3, i_4 = 4, \\ 0, & \text{otherwise} \end{cases}$$

for $a_1, \ldots, a_7, a_{i_1}, \cdots, a_{i_4} \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}^{a_5}_{a_1 a_6}$, $\mathcal{Y}_2 \in \mathcal{V}^{a_6}_{a_2 a_7}$, $\mathcal{Y}_3 \in \mathcal{V}^{a_7}_{a_3 a_4}$, and $w_{(a_{i_k})} \in W^{a_{i_k}}$ with $k = 1, \cdots, 4$.

# Genus-zero Moore-Seiberg equations

The second one is

$$\mathbf{IP}: \quad \coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_6} \otimes \mathcal{V}_{a_6a_7}^{a_5} \otimes \mathcal{V}_{a_3a_4}^{a_7} \longrightarrow (\mathrm{Hom}(W\otimes W\otimes W\otimes W, W))\{x_0, x_2, x_3\}$$

# Genus-zero Moore-Seiberg equations

defined by linearity and by

$$(\mathbf{IP}(\mathcal{Y}_1 \otimes \mathcal{Y}_2 \otimes \mathcal{Y}_3))(w_{(a_{i_1})}, w_{(a_{i_2})}, w_{(a_{i_3})}, w_{(a_{i_4})}; x_0, x_2, x_3)$$
$$= \begin{cases} \mathcal{Y}_2(\mathcal{Y}_1(w_{(a_{i_1})}, x_0)w_{(a_{i_2})}, x_2)\mathcal{Y}_3(w_{(a_{i_3})}, x_3)w_{(a_{i_4})}, & i_1 = 1, i_2 = 2, \\ & i_3 = 3, i_4 = 4, \\ 0, & \text{otherwise} \end{cases}$$

for $a_1, \dots, a_7, a_{i_1}, \cdots, a_{i_4} \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}^{a_6}_{a_1 a_2}$, $\mathcal{Y}_2 \in \mathcal{V}^{a_5}_{a_6 a_7}$, $\mathcal{Y}_3 \in \mathcal{V}^{a_7}_{a_3 a_4}$, and $w_{(a_{i_k})} \in W^{a_{i_k}}$ with $k = 1, \cdots, 4$.

# Genus-zero Moore-Seiberg equations

The third one is

$$\mathbf{I}^{\mathbf{P}}: \quad \coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_7}^{a_6} \otimes \mathcal{V}_{a_2a_3}^{a_7} \otimes \mathcal{V}_{a_6a_4}^{a_5} \longrightarrow (\mathrm{Hom}(W\otimes W\otimes W\otimes W, W))\{y_1, y_2, x_3\}$$

# Genus-zero Moore-Seiberg equations

defined by linearity and by

$$(\mathbf{I}^{\mathbf{P}}(\mathcal{Y}_1 \otimes \mathcal{Y}_2 \otimes \mathcal{Y}_3))(w_{(a_{i_1})}, w_{(a_{i_2})}, w_{(a_{i_3})}, w_{(a_{i_4})}; y_1, y_2, x_3)$$
$$= \begin{cases} \mathcal{Y}_3(\mathcal{Y}_1(w_{(a_{i_1})}, y_1)\mathcal{Y}_2(w_{(a_{i_2})}, y_2)w_{(a_{i_3})}, x_3)w_{(a_{i_4})}, & i_1 = 1, i_2 = 2, \\ & i_3 = 3, i_4 = 4, \\ 0, & \text{otherwise} \end{cases}$$

for $a_1, \dots, a_7, a_{i_1}, \cdots, a_{i_4} \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}^{a_6}_{a_1 a_7}$, $\mathcal{Y}_2 \in \mathcal{V}^{a_7}_{a_2 a_3}$, $\mathcal{Y}_3 \in \mathcal{V}^{a_5}_{a_6 a_4}$, and $w_{(a_{i_k})} \in W^{a_{i_k}}$ with $k = 1, \cdots, 4$.

# Genus-zero Moore-Seiberg equations

The fourth one is

$$\mathbf{II}: \quad \coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_7}\otimes\mathcal{V}_{a_7a_3}^{a_6}\otimes\mathcal{V}_{a_6a_4}^{a_5} \longrightarrow (\mathrm{Hom}(W\otimes W\otimes W\otimes W, W))\{x_0, y_2, x_3\}$$

# Genus-zero Moore-Seiberg equations

defined by linearity and by

$$(\mathbf{II}(\mathcal{Y}_1 \otimes \mathcal{Y}_2 \otimes \mathcal{Y}_3))(w_{(a_{i_1})}, w_{(a_{i_2})}, w_{(a_{i_3})}, w_{(a_{i_4})}; x_0, y_2, x_3)$$
$$= \begin{cases} \mathcal{Y}_3(\mathcal{Y}_2(\mathcal{Y}_1(w_{(a_{i_1})}, x_0)w_{(a_{i_2})}, y_2)w_{(a_{i_3})}, x_3)w_{(a_{i_4})}, & i_1 = 1, i_2 = 2, \\ & i_3 = 3, i_4 = 4, \\ 0, & \text{otherwise} \end{cases}$$

for $a_1, \ldots, a_7, a_{i_1}, \cdots, a_{i_4} \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}^{a_7}_{a_1 a_2}$, $\mathcal{Y}_2 \in \mathcal{V}^{a_6}_{a_7 a_3}$, $\mathcal{Y}_3 \in \mathcal{V}^{a_5}_{a_6 a_4}$, and $w_{(a_{i_k})} \in W^{a_{i_k}}$ with $k = 1, \cdots, 4$.

# Genus-zero Moore-Seiberg equations

The fifth one is

$$\textbf{PI}: \quad \coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_6}^{a_5}\otimes\mathcal{V}_{a_2a_3}^{a_7}\otimes\mathcal{V}_{a_7a_4}^{a_6} \longrightarrow (\mathrm{Hom}(W\otimes W\otimes W\otimes W, W))\{x_1, y_2, x_3\}$$

# Genus-zero Moore-Seiberg equations

defined by linearity and by

$$(\mathbf{PI}(\mathcal{Y}_1 \otimes \mathcal{Y}_2 \otimes \mathcal{Y}_3))(w_{(a_{i_1})}, w_{(a_{i_2})}, w_{(a_{i_3})}, w_{(a_{i_4})}; x_1, y_2, x_3)$$
$$= \begin{cases} \mathcal{Y}_1(w_{(a_{i_1})}, x_1)\mathcal{Y}_3(\mathcal{Y}_2(w_{(a_{i_2})}, y_2)w_{(a_{i_3})}, x_3)w_{(a_{i_4})}, & i_1 = 1, i_2 = 2, \\ & i_3 = 3, i_4 = 4, \\ 0, & \text{otherwise} \end{cases}$$

for $a_1, \dots, a_7, a_{i_1}, \cdots, a_{i_4} \in \mathcal{A}$, $\mathcal{Y}_1 \in \mathcal{V}^{a_5}_{a_1 a_6}$, $\mathcal{Y}_2 \in \mathcal{V}^{a_7}_{a_2 a_3}$, $\mathcal{Y}_3 \in \mathcal{V}^{a_6}_{a_7 a_4}$, and $w_{(a_{i_k})} \in W^{a_{i_k}}$ with $k = 1, \cdots, 4$.

# Genus-zero Moore-Seiberg equations

From the above linear maps, we obtain five isomorphisms.:

$$\mathcal{F}_{12}^{(1)}: \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_6}^{a_5}\otimes\mathcal{V}_{a_2a_7}^{a_6}\otimes\mathcal{V}_{a_3a_4}^{a_7}}{\mathrm{Ker}\ \mathbf{PP}} \longrightarrow \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_6}\otimes\mathcal{V}_{a_6a_7}^{a_5}\otimes\mathcal{V}_{a_3a_4}^{a_7}}{\mathrm{Ker}\ \mathbf{IP}}$$

$$\mathcal{F}_{23}^{(1)}: \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_6}^{a_5}\otimes\mathcal{V}_{a_2a_7}^{a_6}\otimes\mathcal{V}_{a_3a_4}^{a_7}}{\mathrm{Ker}\ \mathbf{PP}} \longrightarrow \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_6}^{a_5}\otimes\mathcal{V}_{a_2a_3}^{a_7}\otimes\mathcal{V}_{a_7a_4}^{a_6}}{\mathrm{Ker}\ \mathbf{PI}}$$

# Genus-zero Moore-Seiberg equations

$$\mathcal{F}_{13}: \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}^{a_5}_{a_1a_6}\otimes\mathcal{V}^{a_7}_{a_2a_3}\otimes\mathcal{V}^{a_6}_{a_7a_4}}{\mathrm{Ker}\ \mathbf{PI}} \longrightarrow \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}^{a_6}_{a_1a_7}\otimes\mathcal{V}^{a_7}_{a_2a_3}\otimes\mathcal{V}^{a_5}_{a_6a_4}}{\mathrm{Ker}\ \mathbf{I}^{\mathbf{P}}}$$

$$\mathcal{F}^{(2)}_{12}: \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}^{a_6}_{a_1a_7}\otimes\mathcal{V}^{a_7}_{a_2a_3}\otimes\mathcal{V}^{a_5}_{a_6a_4}}{\mathrm{Ker}\ \mathbf{I}^{\mathbf{P}}} \longrightarrow \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}^{a_7}_{a_1a_2}\otimes\mathcal{V}^{a_6}_{a_7a_3}\otimes\mathcal{V}^{a_5}_{a_6a_4}}{\mathrm{Ker}\ \mathbf{II}}$$

# Genus-zero Moore-Seiberg equations

$$\mathcal{F}_{23}^{(2)}: \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_6}\otimes\mathcal{V}_{a_6a_7}^{a_5}\otimes\mathcal{V}_{a_3a_4}^{a_7}}{\mathrm{Ker}\ \mathbf{IP}} \longrightarrow \frac{\coprod_{a_1,\cdots,a_7\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_6}\otimes\mathcal{V}_{a_6a_3}^{a_7}\otimes\mathcal{V}_{a_7a_4}^{a_5}}{\mathrm{Ker}\ \mathbf{II}}$$

# Genus-zero Moore-Seiberg equations

**Theorem**

*The above isomorphisms satisfy the following genus-zero Moore-Seiberg equations:*

$$\mathcal{F}_{23}^{(2)} \circ \mathcal{F}_{12}^{(1)} = \mathcal{F}_{12}^{(2)} \circ \mathcal{F}_{13} \circ \mathcal{F}_{23}^{(1)}, \tag{1}$$

$$\mathcal{F} \circ \tilde{\Omega}^{(3)} \circ \mathcal{F} = \tilde{\Omega}^{(1)} \circ \mathcal{F} \circ \tilde{\Omega}^{(4)}, \tag{2}$$

$$\mathcal{F} \circ (\widetilde{\Omega^{-1}})^{(3)} \circ \mathcal{F} = (\widetilde{\Omega^{-1}})^{(1)} \circ \mathcal{F} \circ (\widetilde{\Omega^{-1}})^{(4)}. \tag{3}$$

# On Axiomatic Approaches to Intertwining Operator Algebras

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# $S_3$-symmetry of the Jacobi identity

**Theorem**

*In the presence of the axioms for an intertwining operator algebra except for the associativity property, we assume that there exists an isomorphism*

$$\mathcal{F}: \quad \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4} \otimes \mathcal{V}_{a_2a_3}^{a_5}}{\operatorname{Ker} \mathbf{P}} \longrightarrow \frac{\coprod_{a_1,a_2,a_3,a_4,a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5} \otimes \mathcal{V}_{a_5a_3}^{a_4}}{\operatorname{Ker} \mathbf{I}}$$

*satisfying*

$$\mathcal{F}\left(\pi_P\left(\coprod_{a_5\in\mathcal{A}} \mathcal{V}_{a_1a_5}^{a_4} \otimes \mathcal{V}_{a_2a_3}^{a_5}\right)\right) = \pi_I\left(\coprod_{a_5\in\mathcal{A}} \mathcal{V}_{a_1a_2}^{a_5} \otimes \mathcal{V}_{a_5a_3}^{a_4}\right)$$

# $S_3$-symmetry of the Jacobi identity

**Theorem**

*for any $a_1, \cdots, a_4 \in \mathcal{A}$, and that the Moore-Seiberg equations (2) and (3) hold, then the Jacobi identity for the ordered triple $(\tilde{w}_{(a_1)}, \tilde{w}_{(a_2)}, \tilde{w}_{(a_3)}) \in \coprod_{i=1}^{3} W^{a_i}$ implies the Jacobi identity for the triple $(\tilde{w}_{(a_{\tau(1)})}, \tilde{w}_{(a_{\tau(2)})}, \tilde{w}_{(a_{\tau(3)})}) \in \coprod_{i=1}^{3} W^{a_{\tau(i)}}$ for any $\tau \in S_3$.*

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# Thank You!